Solutions and teaching notes for chapter 6

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 6. *Action:* Revisit the rate $2a + h$.
2. $\mathbb{R} \setminus \{2\}$. *Action:* Set the denominator to zero.
3. Positive outside $[-2, 1]$, negative between the roots, zero at the endpoints. *Action:* Build two sign lines.
4. $x^3 - 2x^2 + x - 2$. *Action:* Distribute each term without omission.
5. They are equal. *Action:* Substitute -x in brackets.
6. $y = 2$. *Action:* Revisit the tangent equation and simplify.

Investigation activity

x^3 is strictly increasing with a horizontal tangent at 0. For a differentiable increasing function, the rates on either side are positive or zero, so the derivative cannot be negative.

Immediate applications

1. $5x^4$ on $\mathbb{R}$; $-\dfrac{2}{x^3}$ for $x \neq 0$; $\dfrac{1}{2\sqrt{x}}$ for $x > 0$.
2. $2x(3x-2) + 3(x^2+1) = 9x^2 - 4x + 3$; $-\dfrac{2x}{(x^2+1)^2}$, on $\mathbb{R}$.
3. $f\prime = 2x - 4$, so decreasing up to 2 then increasing. Minimum -3 at 2; values at the endpoints 1 and -2, so maximum 1 at 0.
4. First even, second odd; difference $x^2 + 1 - 2x = (x-1)^2 \geq 0$, equality at 1.
5. $A(x) = x(6-x)$ on $[0, 6]$; $A\prime = 6 - 2x$. Maximum 9 m^2 at 3.

Exercises: method and results

Exercise 6.1 – A collection of derivatives. $12x^3 - 2$; $2x(x-1) + (x^2+2) = 3x^2 - 2x + 2$; $\dfrac{[2(x+3) - (2x+1)]}{(x+3)^2} = \dfrac{5}{(x+3)^2}$ for $x \neq -3$.

Exercise 6.2 – A complete table. $f\prime = 3(x-1)(x-3)$; signs $+, -, +$. Values $f(0) = 1$, $f(1) = 5$, $f(3) = 1$, $f(4) = 5$. Minimum 1 at 0 and 3; maximum 5 at 1 and 4. The endpoints count.

Exercise 6.3 – A quotient. $f\prime = 1 - \dfrac{4}{x^2} = \dfrac{(x-2)(x+2)}{x^2}$. Sign negative before 2, positive after. Minimum 4 at 2; for $x > 0$, $x + \dfrac{4}{x} \geq 4$, equality only at 2.

Exercise 6.4 – Comparing two curves. $f - g = x(x-1)(x+1)$. Signs: negative on $(-\infty,-1)$, positive on $(-1,0)$, negative on $(0,1)$, positive on $(1,+\infty)$. Intersection at $-1,0,1$. We can make the table of the three factors without differentiating.

Exercise 6.5 – A box. $V = x(12-2x)^2 = 4x(6-x)^2$. $V\prime = 12(x-2)(x-6)$, positive on $[0,2)$, negative on $(2,6)$. $V(0) = V(6) = 0$, $V(2) = 128$ cm^3. Maximum for x=2 cm; box $8 \times 8 \times 2$.

Exercise 6.6 – Stationary without an extremum. $f\prime = 3x^2$, so value 0 at 0. In every neighbourhood of 0, negative x give $f(x) < 0$ and positive x give $f(x) > 0$; no extremum.

Exercise 6.7 – Comparing strategies. Square: $(x-3)^2 + 3 \geq 3$. Derivative: $2x - 6$, minimum at 3, value 3. The first avoids a complete study of variation; the second illustrates the link between chapters.

Exercise 6.8 – Delayed review. It decreases on $(-\infty,0)$ and $(0,+\infty)$. It does not decrease on their union: $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$. The domain is not an interval.

CQFD, exercise 12 p. 156, questions 1 and 2. $f(x) - g(x) = \frac{1}{2}x^2 - 1 = \frac{1}{2}(x-\sqrt{2})(x+\sqrt{2})$. The curve of f is strictly above that of g for $x < -\sqrt{2}$ or $x > \sqrt{2}$, below between these two numbers; equality at the x-coordinates $\pm\sqrt{2}$.

CQFD, exercise 16 p. 156, questions f and g. For $f(x) = x^3$, we can choose $(-\infty,0)$: $f(x) < 0$ and the function is strictly increasing. For $g(x) = x^2 - 4$, we can choose $[0,2)$: $g(x) < 0$ and the function is strictly increasing on this interval. Subintervals that are not reduced to a point also work.

MCQ: answers and reasons

MCQ 1.

1. B. Integer power.
2. A. Rate of the reciprocal.
3. D. Product rule.
4. C. Sign of the derivative and variation.
5. B. Nonzero denominator.

MCQ 2.

1. C. $f(-x) = -f(x)$.
2. A. The other conclusions require more.
3. D. Formula and domain.
4. B. Reached at 1.
5. C. Derivative $2x_0 = 0$.

True or false

1. False: $f(x) = \dfrac{1}{x} > 0$ on $(0, +\infty)$, but $f\prime(x) < 0$.
2. True, with the hypotheses stated; at the endpoints, the conclusion is not imposed.
3. False without sign hypotheses: on $[-2, -1]$, $u(x) = v(x) = x$ are increasing but x^2 is decreasing.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.